<Technical Solution Name>

In partial fulfillment of the requirements for the course
Visualizations and Data Analysis
1st Semester, S.Y. 2026-2027

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October 2026

Table of Contents

<Technical Solution Name>.....1
Table of Contents.....2
I. Problem Background and Validation.....3
II. The Project.....3
III. Output.....4
IV. Conclusion.....4
V. Development Team.....4
VI. References.....6

I. Problem Background and Validation

Understanding the structural composition of the Philippines' solid waste crisis and energy grid vulnerabilities is crucial for knowing and evaluating how municipal waste generation, surging energy costs, and depleting natural resources impact everyday citizens and industrial stability of the Philippines.

a. The Problem

In the Philippines, municipalities are having a difficult time dealing with the municipal wastes that produce everyday, while having to manage the energy cost that the residents have to shoulder every single month just to live a modern life. In South East Asia, the Philippines is one of the countries with the highest cost of electricity and one of the main sources of waste pollution in the world. Everyone is struggling with maintaining and disposing and properly segregating our wastes, leading to a severe environmental degradation and overflowing landfills. Furthermore, Filipino consumers face heavily fluctuating and expensive power costs (such as Meralco residential and commercial rates), making electricity a heavy burden for households alongside the country's heavy reliance on imported fossil fuels and depleting natural resources to sustain the grid. The most affected are Filipino households and consumers who bear the brunt of expensive utility bills and environmental pollution, as well as Local Government Units (LGUs), Energy Policymakers, and Utility Providers who need these insights to implement sustainable, alternative baseload generation strategies like Waste-to-Energy (WtE). The existence of this dual waste and energy crisis is verified by official national records from the Philippine Statistics Authority (PSA), the Department of Environment and Natural Resources (DENR) - National Solid Waste Management Commission (NSWMC), and energy regulatory reports highlighting domestic resource depletion and power generation costs, alongside rising utility statements and garbage management data that continuously emphasize the urgency of transitioning toward alternative energy solutions.

The most affected are Filipino households and consumers who bear the brunt of expensive utility bills and environmental pollution. Additionally, Local Government Units (LGUs), Energy Policymakers, and Utility Providers need these insights to implement sustainable, alternative baseload generation strategies like Waste-to-Energy (WtE).

b. Problem Validation

The dual crisis of waste and energy in the Philippines is not a concern as it is truly proven and is happening today. Official records from the Department of Environment and Natural Resources (DENR) and the National Solid Waste Management Commission (NSWMC) show how much pressure municipal solid waste places on the country’s landfills. Landfills are filling up fast. Waste is piling up. The data does not lie.

At the time energy reports from regulators and real bills from consumers show that electricity prices remain high.. They keep changing. One month it's one rate. The next it’s another. The cost of power is not stable. It is unpredictable. Companies like Meralco confirm this every time a bill arrives.

More troubling economic data from the Philippine Statistics Authority (PSA) paints a picture of real people struggling. The Family Income and Expenditure Survey (FIES) shows how families are being squeezed. More money goes to paying for electricity. More money goes to waste collection. Less money remains for food, school, healthcare. The double burden hits hardest, at home.

The evidence is clear. The waste and energy crisis is real. It is growing and it is affecting the lives of everyone .

II. The Project

This project is designed to evaluate the feasibility and the impact of the use of Waste to Energy Systems in both the environmental and economical economics. By analyzing the waste amount or volumes, the power generation costs, and international data, this project aims to show how Waste to Energy can simultaneously decrease and help with the mitigation of pollution, in lowering the energy costs, and substitute for the natural resources that we used to generate electricity such as fossil fuels.

a. Dataset

The datasets originate from official environmental statistics, energy regulatory boards, and international database repositories tracking global WtE growth and comparative pricing.

The empirical analysis of this project relies on three primary data subsets, mostly pertaining to, Philippine Energy Metrics, National Solid Waste Records, and International Waste to Energy Benchmarking Data.

The first dataset shows the Philippines energy consumption and pricing trends over the course of 12 years from 2012 to 2024, and the primary attributes that we are tracking are the observation year, grid region, total consumption and sales in gigawatt/ hour, which is mostly strings and floats and integers.

The second dataset shows the Municipal Solid Waste generation, over the 80 to 100 regional and national entries across multi year reporting periods. The key parameters that we get from this is the regional classification, annual and daily waste generation, and the breakdown of what categories the wastes are.

Finally, we have the comparative dataset benchmarking of Singapore's WtE infrastructure that has been compiled spanning over 15 annual summary records, this data records total waste generated versus the incinerated, the landfill diversion rates, the total electricity produced, and these parameters are mostly integers, floats and strings.

The data covers multi-year metrics including total municipal solid waste generation by classification, historical kilowatt-hour (kWh) utility rates, local resource reserve estimates, and international WtE capacity outputs.

To prepare the datasets, several cleaning steps and transformations will be applied. Missing entries will be removed, anomalies or outliers in waste generation reports will be cross-referenced with DENR/PSA reports, and categorical variables (such as regional classifications and waste types) will be standardized into numerical formats to ensure seamless correlation and analysis.

b. Data Dashboard

<Explain what the dashboard is for e.g., visualization, decision support, etc>

The dashboard is created to visualize the increasing rate of electricity in the Philippines and the comparison with the likes of Singapore who uses a Waste To Energy Facilities to convert their waste into electricity. There are also comparisons on how much waste pollution is produced and left in the landfills between the two countries. Lastly, this dashboard is to show the parameters and variables that affect both energy graph and pollution over time and what could be the result if we were to use a Waste to Energy Facilities here in the Philippines.

<List main dashboard components and functionalities>

c. Project Description

<Briefly describe the overall project or system>

<List the key objectives or deliverables>

III. Output

<Insert screenshots, charts, tables, or figures showing the final product.>



Figure 1-1. Total electricity consumption in the Philippines from 2012 to 2024

The analysis shows that electricity consumption in the Philippines generally increased from 2012 to 2024. Luzon had the highest consumption, followed by Visayas and Mindanao. The total consumption increased from 72.92 TWh in 2012 to 126.94 TWh in 2024, representing an increase of approximately 74.1%. There was a decrease in total consumption in 2020 compared with 2019, but consumption increased again in the following years. Overall, the results show the growing demand for electricity in the Philippines and highlight the importance of exploring sustainable energy sources, such as waste-to-energy, to help meet future energy needs.

<Summarize the most important outcomes of your project or analysis>
<Add storyboard panels that show before/after scenarios or end-user experience>

IV. Conclusion

<Provide a paragraph summarizing the project's impact and expected outcomes>

V. Development Team

The group members have elected Mr. Roy Mendoza, as its own leader who leads and designates the tasks for this project. With his specialty in data analysis and data finding for what is needed, and the specialty of Mr. Ryo Mendoza, in coding and also analysis, along with Mr. Obias ability to provide guidance when the group is overthinking, and ability to deeply analyze some parts, this group is a balanced one for the creation of this project.

a. Team Description
<List all members with their roles>

NAME OF MEMBER	ROLES	DESCRIPTION
 **Roy Gabriel I. Mendoza**	● Leader ● Dataset finder ● Data set cleaner ● Presenter ● Documenter ● Tester	The goal of this student is to lead the group and give the initial insights on what to do and how to do it. The student would lead the group by giving them tasks and dates to finish them by to properly align the work with the dates and schedules available for everyone.
BACKGROUND This student's background from way back in highschool is one with research focus one, since grade 7 there are already research focus goals in subjects.	**EXPERTISE** ● Researching ● Data Finding ● Data Set segregation ● Presentation	**CONTRIBUTIONS/RESPONSIBILITIES** 1. Looking for the necessary data sets 2. Analyzing what data variables must be kept and removed 3. Presenting 4. Code Testing
 **Ryo Shin P. Mendoza**	● Dataset Analyzer ● Graph Coder ● Presenter ● Documenter ● Tester	The goal of this student is to analyze the dataset that the other members would find and see if they are viable to be used and to code them after being cleaned to show them into a graph and implement it into the dashboard. He is also tasked with the code creation of the dashboard since he is the best of the members to create the data dashboard.
BACKGROUND ●	**EXPERTISE** ●	**CONTRIBUTIONS/RESPONSIBILITIES** 1.

 **Jave Joaquin C. Obias**	• Dataset Analyzer • Tester • Presenter • Documenter	The goal of this student is also to analyze the data and files that other students would use and create along with leading with the documentation of all work that is done to make the leader's calendar dues with the group follow the initial plan.
BACKGROUND •	**EXPERTISE** •	**CONTRIBUTIONS/RESPONSIBILITIES** 1.

<Brief paragraph describing the team's dynamics, expertise, and goals>

b. Profile
<Provide a short bio or background for each member, including skills, expertise, and contributions>

c. Roles and Responsibilities
<Describe who is responsible for which task, e.g., data collection, system design, documentation>

VI. References

Philippines Energy Consumption:

https://www.statista.com/statistics/1267088/philippines-electricity-sales-and-consumption-by-grid/

ps://docs.google.com/spreadsheets/d/1oGc1iGpacELS6tkHjesvAeKZo8v2iv4UkYFY04e0p-I/edit?gid=441532959#gid=441532959

https://doe.gov.ph/data-and-prices/electricity

Philippines Pollution Statistics:

Waste: https://datacommons.org/explore?#q=philippines%20wastes

Air: https://www.kaggle.com/datasets/shruthiiiee/asia-2023-air-report-2000-cities

Other countries energy cost that uses the WtE:

Singapore: https://tablebuilder.singstat.gov.sg/table/TS/M890991

Reduction in Pollution with the use of WtE:

Singapore: https://www.sg101.gov.sg/environment/case-studies/wm/

NOTE:
If adding screenshots or images in your document, you can do the following.
1.

"SCREENSHOT OR IMAGE"

Figure 1: <Name of the Figure (e.g. Bar Graph for the Problem Validation)>

<Add its description here> e.g. "Figure 1 presents/illustrates/shows the following … "

2.

Table 1: <Name of the Table>

Screenshot or Image	Description
"SCREENSHOT OR IMAGE"	**"DESCRIPTION OF THE IMAGE"**

<Add its description here> e.g. "Table 1 shows the following … "